

VYBORG, Russian Federation

WMO: 228920

Lat: 60.717N

Lon: 28.733E

Elev: 13

StdP: 101.17

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.7	-20.8	-26.4	0.3	-23.7	-23.3	0.5	-20.7	12.4	-0.2	11.5	-0.2	1.1	0	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	27.2	19.5	25.3	18.5	23.6	17.6	20.7	25.3	19.6	24.0	18.5	22.5	3.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	23.4	17.8	12.8	22.1	16.8	12.0	20.9	59.6	25.3	55.9	24.0	52.4	22.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.4	8.2	DB	-25.6	29.3	4.4	2.2	-28.8	30.9	-31.3	32.2	-33.8	33.4	-37.0	35.1	(4)
(5)			WB	-25.6	21.8	4.3	1.4	-28.7	22.8	-31.3	23.7	-33.7	24.4	-36.8	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-6.4	-6.8	-2.9	3.3	10.3	15.2	18.6	16.8	11.7	5.6	0.5	-3.4	(6)	
	DBStd	9.88	6.75	6.66	4.55	3.63	4.14	3.50	3.18	2.97	3.27	3.75	4.20	5.86	(7)	
	HDD10.0	2493	509	469	401	203	47	2	0	0	21	141	284	416	(8)	
	HDD18.3	4841	767	702	659	451	252	106	36	66	199	394	534	674	(9)	
	CDD10.0	770	0	0	0	2	56	159	266	211	72	5	0	0	(10)	
	CDD18.3	74	0	0	0	0	2	12	43	18	0	0	0	0	(11)	
	CDH23.3	429	0	0	0	0	12	76	253	88	0	0	0	0	(12)	
	CDH26.7	73	0	0	0	0	1	12	49	12	0	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.4	3.3	3.2	2.9	3.3	3.4	3.1	3.0	3.2	3.5	3.7	3.9	(14)	
(15) Precipitation	PrecAvg	693	46	35	43	37	39	52	70	81	80	73	75	65	(15)	
	PrecMax	1014	76	89	99	97	88	119	146	189	159	183	176	126	(16)	
	PrecMin	486	7	1	11	4	6	2	18	17	25	20	6	14	(17)	
	PrecStd	138	19	18	22	21	21	32	35	41	38	43	31	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	3.7	7.9	17.6	24.8	28.2	30.3	28.2	21.5	14.1	9.1	7.2	(19)	
		MCWB	2.7	1.9	3.1	10.7	16.6	19.3	21.0	19.9	16.2	11.9	7.9	6.2	(20)	
	2%	DB	2.5	2.3	5.1	14.0	21.9	25.5	27.9	25.7	19.3	12.8	7.9	5.2	(21)	
		MCWB	1.9	1.3	1.9	8.7	14.6	18.1	20.1	18.8	15.1	11.4	7.0	4.5	(22)	
	5%	DB	1.7	1.6	3.2	11.2	19.4	23.2	26.1	23.8	17.9	11.8	7.0	4.1	(23)	
		MCWB	1.2	0.8	1.1	7.0	13.0	16.8	19.4	17.9	14.6	10.6	6.3	3.4	(24)	
	10%	DB	1.0	1.0	2.1	9.1	17.2	21.2	24.3	22.1	16.6	10.8	5.9	3.0	(25)	
		MCWB	0.5	0.4	0.7	5.3	11.8	15.6	18.3	17.2	13.8	9.7	5.2	2.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.5	4.0	11.6	17.8	21.0	22.5	21.5	17.3	12.7	8.2	6.2	(27)	
		MCDB	3.2	3.3	6.5	16.1	22.4	26.0	28.2	25.3	19.9	13.5	8.7	7.0	(28)	
	2%	WB	2.0	1.6	2.5	9.3	15.5	19.5	21.2	19.7	16.1	11.7	7.1	4.7	(29)	
		MCDB	2.3	2.1	4.1	13.4	20.3	23.9	26.1	24.1	18.1	12.5	7.7	5.1	(30)	
	5%	WB	1.3	1.0	1.6	7.2	14.0	17.8	20.1	18.8	15.1	10.7	6.3	3.5	(31)	
		MCDB	1.6	1.4	2.7	10.4	18.6	21.6	24.6	22.8	17.2	11.6	6.9	4.0	(32)	
	10%	WB	0.6	0.4	1.0	5.8	12.3	16.4	19.1	17.8	14.3	9.9	5.3	2.5	(33)	
		MCDB	1.0	0.9	1.8	8.5	16.3	20.2	23.3	21.1	16.1	10.7	5.8	2.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.7	6.6	7.0	7.7	6.9	7.0	7.0	6.0	4.2	3.4	4.1	(35)	
		MCDBR	3.2	4.0	7.0	11.2	11.2	10.2	9.8	9.7	7.6	4.6	3.6	3.2	(36)	
	5% WB	MCWBR	3.2	3.6	5.0	6.7	5.8	4.7	4.0	4.4	4.5	3.6	3.6	3.1	(37)	
		MCWBR	3.2	3.8	5.4	10.0	9.8	8.5	8.4	8.2	6.4	4.0	3.3	3.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.267	0.306	0.329	0.368	0.364	0.362	0.387	0.376	0.350	0.332	0.294	0.247	(40)	
		taud	2.089	2.122	2.181	2.296	2.414	2.457	2.406	2.429	2.490	2.385	2.180	2.042	(41)	
	Edn at Noon		457	658	781	824	861	868	832	812	773	645	424	316	(42)	
		Edh at Noon	42	75	98	107	103	100	103	93	74	59	39	27	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.25	0.82	2.28	3.73	5.13	5.37	5.26	4.06	2.40	0.96	0.26	0.13	(44)	
	RadStd		0.05	0.18	0.29	0.41	0.59	0.36	0.49	0.45	0.25	0.13	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	-183	-189	N/A	N/A

Nomenclature: See separate page